

Discussion 2 Worksheet

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Problem 1:

- (i) Fill in the table

p	q	$\neg p$	$\neg q$	$p \wedge q$	$\neg p \vee \neg q$	$\neg q \wedge \neg p$	$p \Rightarrow q$	$\neg p \Rightarrow \neg q$	$\neg q \Rightarrow p$
T	T								
T	F								
F	T								
F	F								

- (ii) Identify if the following statements are true using the truth table above (extend the truth table if necessary)

(a) $(p \wedge q) \Leftrightarrow \neg(\neg p \vee \neg q)$

(b) $(p \Rightarrow q) \Leftrightarrow (\neg p \Rightarrow \neg q)$

(c) $\neg(\neg q \Rightarrow p) \Leftrightarrow (\neg q \wedge \neg p)$

Problem 2: Determine the logical negation of each claim.

- (i) There exists a country where every city has a university.
- (ii) For all positive real numbers x , $-x < 0$.
- (iii) $\exists x \in \mathbb{R} : \forall y, z \in \mathbb{Z}, \neg P(x, y, z)$
- (iv) $\forall x, y \in \mathbb{R} : (x < y \Rightarrow \exists q \in \mathbb{Q} : x < q < y)$

Problem 3: Write the negation, converse, inverse, and contrapositive of each of the following statements.

- (i) Every polygon that is a square has four sides.
- (a) Negation:
- (b) Converse:
- (c) Inverse:

(d) Contrapositive:

(ii) $\forall x \in \mathbb{R}, (x > 2 \Rightarrow x^2 > 4)$

(a) Negation:

(b) Converse:

(c) Inverse:

(d) Contrapositive:

Problem 4: For each of the following, determine whether P and Q are logically equivalent (i.e., whether $P \Leftrightarrow Q$ is true).

(a) Given

$$P : 2x - 3 = 7, \quad Q : x > 4$$

(b) Let X and Y be arbitrary sets. Given

$$P : X \subseteq Y, \quad Q : X \cup Y = Y$$

Problem 5: Define perfect squares using Biconditional statement.